

TITLE

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1. CONDENSED SETS

Definition 1.0.1. A *profinite set* is a set which is the projective limit of a (small) directed system of finite sets.

A profinite set $X = \varprojlim_i X_i$ has a natural topology, referred to as the *profinite topology* or *limit topology*. It can be formulated in the following equivalent ways:

(♠ TODO: define inverse limit topology)

1. As the inverse limit topology with respect to the family of projection maps $\pi_i : X \rightarrow X_i$ onto the finite sets X_i , each equipped with the discrete topology, in the system.
2. As the subspace topology on $\varprojlim_i X_i$ inherited from the product topology on $\prod_i X_i$.
3. As the topology generated by the basis of "finite index" sets (specifically, the clopen sets) of the form $\pi_i^{-1}(V)$ for some index i and some subset $V \subseteq X_i$.

These topologies do not depend on the inverse system representing X .

Proposition 1.0.2. Let X be a topological space. The following conditions are equivalent:

- X is a profinite set (Definition 1.0.1).
- X is compact, Hausdorff, and totally disconnected.
- X is a Stone space.

Definition 1.0.3. Let S and T be profinite sets (Definition 1.0.1). A *morphism $S \rightarrow T$ of profinite sets S and T* can be characterized in the following equivalent ways:

1. A continuous map between the compact Hausdorff totally disconnected topological spaces.
2. ... (♠ TODO: how to characterize directly on the inverse limit presentation)

Definition 1.0.4. The *category of profinite sets*, denoted by notations such as $\mathbf{Prof}$, is the category whose objects are profinite sets (Definition 1.0.1) and whose morphisms are morphisms of profinite sets (Definition 1.0.3).

This category is equivalent to the category of compact Hausdorff totally disconnected topological spaces whose morphisms are continuous maps.

This category is also equivalent to the pro-category of the category of finite sets (♠ TODO: category of finite sets).

$\mathbf{Prof}$ admits all small limits.

Definition 1.0.5 ([?, Definition 2.2.], [?, Definition 1.1.]). Define the following Grothendieck topology on the category $\mathbf{Prof}$ (Definition 1.0.1) of profinite sets (Definition 1.0.1): a finite collection $(f_i : S_i \rightarrow S)_{i \in I}$ of maps in $\mathbf{Prof}$ is a covering if the maps are jointly surjective, i.e. $\coprod_{i \in I} S_i \rightarrow S$ is surjective.

One may also call $\mathbf{Prof}$ equipped with this Grothendieck topology the *pro-étale site of the point*, and denote this site by $*_{\mathbf{pro\acute{e}t}}$.

It should be noted that $\mathbf{Prof}$ is not essentially small. (♠ TODO: talk about how the grothendieck topology here is finitary)

Definition 1.0.6 ([?, Definition 2.3.]). Let κ be a cardinal. A profinite set (Definition 1.0.1) S is called *κ -small* if the set of clopen subsets of S has cardinality $< \kappa$. The full subcategory of κ -small profinite sets is denoted $\mathbf{Prof}_\kappa \subset \mathbf{Prof}$ (Definition 1.0.4), and this category is equipped with the Grothendieck topology which is the restriction of the one (Definition 1.0.5) on $\mathbf{Prof}$ to $\mathbf{Prof}_\kappa$. (♠ TODO: restriction of a site to a subcategory).

It should be noted that $\mathbf{Prof}_\kappa$ is an essentially small site, unlike $\mathbf{Prof}$.

Definition 1.0.7 ([?, Definition 2.4.]). Let κ be a cardinal. A *κ -condensed set* is a sheaf of sets on $\mathbf{Prof}_\kappa$ (Definition 1.0.6).

The category of κ -condensed sets is denoted by $\mathbf{CondSet}_\kappa$.

Explicitly, a κ -condensed set X is a functor

$$X : \mathbf{Prof}_\kappa^{op} \rightarrow \mathbf{Set}$$

such that

1. If $(S_i)_{i \in I}$ is a finite collection of objects of $\mathbf{Prof}_\kappa^{op}$, then

$$X(\sqcup_{i \in I} S_i) \xrightarrow{\sim} \prod_{i \in I} X(S_i);$$

2. If $f : T \rightarrow S$ is a surjective map in $\mathbf{Prof}_\kappa$, then

$$X(S) \xrightarrow{f^*} X(T)$$

is injective with image equal to the set of those $x \in X(T)$ for which $p_1^*x = p_2^*x$, where $p_1, p_2 : T \times_S T \rightarrow T$ are the two projections.

(♠ TODO: talk about how the category of extremely disconnected sets can be used to alternatively define condensed sets.)

Definition 1.0.8. Let $\mathcal{A}$ be a category. Let κ be a cardinal

A κ -condensed object of $\mathcal{A}$ is a sheaf on Prof_κ (Definition 1.0.8) valued in $\mathcal{A}$.

Names for objects of $\mathcal{A}$ induce names for κ -condensed objects of $\mathcal{A}$, e.g. a κ -condensed group is a κ -condensed object of the category of groups, etc.

As per Proposition A.0.1, for γ a “kind of algebraic structure defined by finite projective limits”, the category of κ -condensed objects of γ – **Sets** is equivalent to the category of γ -objects of CondSet_κ (Definition 1.0.7).

Definition 1.0.9 ([?, After Definition 2.4.]). The *category of condensed sets*, denoted CondSet , is the colimit category (♠ TODO: define limit/colimit categories)

$$\text{CondSet} := \varinjlim_{\kappa} \text{CondSet}_\kappa$$

(Definition 1.0.6) over all cardinals κ , where the functors $\text{CondSet}_\kappa \rightarrow \text{CondSet}_{\kappa'}$ for $\kappa < \kappa'$ are the pullback functors of the inclusion functors $\text{Prof}_\kappa \hookrightarrow \text{Prof}_{\kappa'}$. (♠ TODO: pullback functors)

This category is generated under colimits by arbitrary profinite sets (♠ TODO: explain more on the yoneda embedding)

Definition 1.0.10. Let $\mathcal{A}$ be a category.

A *condensed object of $\mathcal{A}$* is an object (♠ TODO: the intent here is for us to not say κ .)

The category of condensed objects of $\mathcal{A}$ may be denoted as $\text{Cond}(\mathcal{A})$. For example, $\text{Cond}(\text{Ab})$ denotes the category of condensed abelian groups .

Definition 1.0.11. Given a topological space T , define its *associated condensed set* $\underline{T}$ as the functor (♠ TODO: technically, do I define it on Prof_κ or Prof ?)

$$\text{Prof}^{\text{op}} \rightarrow \mathbf{Sets}, \quad S \mapsto C(S, T),$$

where $C(S, T)$ is the set of continuous maps from S to T . (♠ TODO: talk about how the yoneda embedding of profinite sets into the category of condensed sets is given in the same way.)

Note that if T has the structure of a topological group, topological ring, etc., then $\underline{T}$ is condensed group (Definition 1.0.10), condensed ring, etc.

2. SOLID CONDENSED ABELIAN GROUPS

Definition 2.0.1 ([?, Before Definition 2.4.]). Let $S = \varprojlim_i S_i$ be a profinite set (Definition 1.0.1) with S_i finite discrete sets.

Define

$$\mathbb{Z}[S]^\blacksquare := \varprojlim \mathbb{Z}[S_i]$$

where $\mathbb{Z}[S_i]$ is the free ring generated by S_i and the morphisms among these is the natural morphism between the free rings induced by the morphisms of finite sets. It is appropriate to call $\mathbb{Z}[S]^\blacksquare$ the *solidification of the free ring $\mathbb{Z}[S]$ as a condensed abelian group* (Definition 1.0.10) — indeed, see (♠ TODO:)

Definition 2.0.2 ([?, Definition 2.4.]). A condensed abelian group (Definition 1.0.10) is solid if for any profinite set (Definition 1.0.1) S with a map $f : S \rightarrow M$, there is a unique map $\tilde{f} : \mathbb{Z}[S]^\blacksquare \rightarrow M$ (Definition 2.0.1) such that the composition (♠ TODO: talk about yoneda embedding so that S makes sense as a condensed set)

$$S \rightarrow \mathbb{Z}[S]^\blacksquare \xrightarrow{\tilde{f}} M$$

is f .

The full subcategory of $\text{Cond}(\text{Ab})$ (Definition 1.0.10) consisting of solid objects may be denoted by Solid or Sol .

Theorem 2.0.3 ([?, Theorem 5.8.(i), Theorem 6.2.(i)], see also [?, Theorem 2.5.]).

1. The category Solid of solid abelian groups is closed under small limits, small colimits, and extensions as a subcategory of the category $\text{Cond}(\text{Ab})$ (Definition 1.0.10).
2. The inclusion functor $\text{Solid} \hookrightarrow \text{Cond}(\text{Ab})$ admits a left adjoint, denoted by $M \mapsto M^\bullet$ and called *solidification*. As an endofunctor of $\text{Cond}(\text{Ab})$, the solidification functor commutes with all colimits and takes $\mathbb{Z}[S]$ to $\mathbb{Z}[S]^\bullet$ (Definition 2.0.1).
3. The category of solid abelian groups admits compact projective generators, which are exactly the condensed abelian groups of the form $\prod_I \mathbb{Z}$ for sets I .
4. There is a unique symmetric monoidal tensor product, denoted by $\otimes^\blacksquare$, on Solid , making the solidification functor $\text{Cond}(\text{Ab}) \rightarrow \text{Solid} : M \mapsto M^\bullet$ symmetric monoidal. (♠ TODO: what is the symmetric monoidal functor on CondAb then?)

3. SOLID MODULES

Definition 3.0.1. Let R be a topological ring.

A *solid R -module* is a solid condensed abelian group (Definition 2.0.2) that is also a module of $\underline{R}$ (Definition 1.0.11) (♠ TODO: should have to think — given a condensed abelian group, what site is it a sheaf on, and what is underline a sheaf on?)

(♠ TODO: what are morphisms)

The category of solid R -modules may be denoted by Solid_R or Sol_R .

Given two topological rings R and S , we may accordingly define the notion of *solid R - S -bi-module*. The category of such objects may be denoted by ${}_R\mathrm{Solid}_S$ or ${}_R\mathrm{Sol}_S$.

(♠ TODO: solid R -sheaf on a scheme)

(♠ TODO: proetale site) (♠ TODO: frobenius operator on)

APPENDIX A. MISCELLANEOUS

Proposition A.0.1 ([GV72, Exposé II, Proposition 6.3.1]). Let $\mathcal{C}$ be a site, and let γ be a “kind of algebraic structure defined by finite projective limits” (♠ TODO: Probably the correct terminology for this is “essentially algebraic categories” or “Lawvere theories”) (e.g. γ can be the notion of group objects or abelian group objects, ring objects, or module objects). The following categories are equivalent:

1. The category of sheaves on $\mathcal{C}$ with values in the category of γ -sets,
2. The category of γ -objects of the category of sheaves of sets on $\mathcal{C}$, and
3. The category of γ -objects of the category of presheaves of sets on $\mathcal{C}$ whose underlying presheaf is a sheaf of sets.

REFERENCES

- [AP07] Jeffrey D. Achter and Rachel Pries. The integral monodromy of hyperelliptic and trielliptic curves. *Mathematische Annalen*, 338(1):187–206, 2007.
- [Arm92] Brumer Armand. The average rank of elliptic curves i. *Inventiones mathematicae*, 109(1):445–472, 1992.
- [Ayo23] Joseph Ayoub. Counterexamples to F. Morel’s conjecture on $\pi_0^{\mathfrak{D}^1}$. *Comptes Rendus. Mathématique*, 361:1087–1090, 2023.
- [Bac24] Tom Bachmann. Strongly a1-invariant sheaves (after f. morel), 2024.
- [Bal04] Paul Balmer. The spectrum of prime ideals in tensor triangulated categories, 2004.
- [BBD82] Alexander A. Beilinson, Joseph Bernstein, and Pierre Deligne. Analyse et topologie sur les espaces singuliers (i). *Astérisque*, 100, 1982.
- [BBDG18] Alexander A. Beilinson, Joseph Bernstein, Pierre Deligne, and Ofer Gabber. *Faisceaux pervers*, volume 4. Société mathématique de France Paris, 2018.
- [BBK⁺23] Barinder S. Banwait, Armand Brumer, Hyun Jong Kim, Zev Klagsbrun, Jacob Mayle, Padmavathi Srinivasan, and Isabel Vogt. Computing nonsurjective primes associated to galois representations of genus 2 curves. *LuCaNT: LMFDB, Computation, and Number Theory*, 796:129, 2023.
- [BC19] Tilman Bauer and Magnus Carlson. Tensor products of affine and formal abelian groups. *Documenta Mathematica*, 24:2525–2582, 2019.
- [BFK⁺17] Valentin Blomer, Étienne Fouvry, Emmanuel Kowalski, Philippe Michel, and Djordje Milićević. Some applications of smooth bilinear forms with kloosterman sums. *Proceedings of the Steklov Institute of Mathematics*, 296:18–29, 2017.
- [BGI71] Pierre Berthelot, Alexander Grothendieck, and Luc Illusie. *Théorie des Intersections et Théorème de Riemann-Roch (SGA6)*, volume 225 of *Lecture Notes in Mathematics*. Springer-Verlag, 1971.
- [BH12] Salma Baig and Chris Hall. Experimental data for goldfeld’s conjecture over function fields. *Experimental Mathematics*, 21(4):362–374, 2012.

- [BLGHT11] Tom Barnet-Lamb, David Geraghty, Michael Harris, and Richard Taylor. A family of Calabi–Yau varieties and potential automorphy ii. *Publications of the Research Institute for Mathematical Sciences*, 47(1):29–98, 2011.
- [Bor12] Armand Borel. *Linear algebraic groups*, volume 126 of *Graduate Texts in Mathematics*. Springer New York, 2012.
- [BS15a] Manjul Bhargava and Arul Shankar. Binary quartic forms having bounded invariants, and the boundedness of the average rank of elliptic curves. *Annals of Mathematics*, pages 191–242, 2015.
- [BS15b] Manjul Bhargava and Arul Shankar. Ternary cubic forms having bounded invariants, and the existence of a positive proportion of elliptic curves having rank 0. *Annals of Mathematics*, pages 587–621, 2015.
- [BS15c] Bhargav Bhatt and Peter Scholze. The pro-étale topology for schemes. *Astérisque*, 360:99–201, 2015.
- [BSD65] Bryan John Birch and Peter Francis Swinnerton-Dyer. Notes on elliptic curves. ii. *Journal für die reine und angewandte Mathematik*, 218:79–108, 1965.
- [BSS18] Bhargav Bhatt, Christian Schnell, and Peter Scholze. Vanishing theorems for perverse sheaves on abelian varieties, revisited. *Selecta Mathematica*, 24:63–84, 2018.
- [Cho08] Utsav Choudhury. Homotopy theory of schemes and a^1 -fundamental groups. Master’s thesis, Università degli Studi di Padova, 2008.
- [CHT08] Laurent Clozel, Michael Harris, and Richard Taylor. Automorphy for some l -adic lifts of automorphic mod l galois representations. *Publications mathématiques*, 108:1–181, 2008.
- [DA73] Pierre Deligne and Michael Artin. *Théorie des Topos et Cohomologies Étale des Schémas. Séminaire de Géométrie Algébrique due Bois-Marie 1963-1964 (SGA 4)*. Lecture Notes in Mathematics. Springer Berlin, 1973.
- [DBG⁺77] Pierre Deligne, Jean-François Boutot, Alexander Grothendieck, Luc Illusie, and Jean-Louis Verdier. *Étale Cohomology. Séminaire de Géométrie Algébrique due Bois-Marie 1963-1964 (SGA 4 1/2)*. Lecture Notes in Mathematics. Springer-Verlag, 1977.
- [Del80] Pierre Deligne. La conjecture de Weil : II. *Publications Mathématiques de l’IHÉS*, 52:137–252, 1980.
- [Del89] Pierre Deligne. Le groupe fondamental de la droite projective moins trois points. In *Galois Groups over $\mathbb{Q}$: Proceedings of a Workshop Held March 23–27, 1987*, pages 79–297. Springer, 1989.
- [Die02] Luis V. Dieulefait. Explicit determination of the images of the Galois representations attached to abelian surfaces with $\text{End}(A) = \mathbb{Z}$. *Experiment. Math.*, 11(4):503–512 (2003), 2002.
- [DR04] Luis V. Dieulefait and Victor Rotger. The arithmetic of qm -abelian surfaces through their galois representations. *Journal of Algebra*, 281:124–143, 2004.
- [Dri89] Vladimir Gershonovich Drinfeld. Cohomology of compactified manifolds of modules of f -sheaves. *Journal of Soviet Mathematics*, 46(2):1789–1821, 1989.
- [Dru22] Anderi Eduardovich Druzhinin. Stable A^1 -connectivity over a base. *Journal für die reine und angewandte Mathematik (Crelles Journal)*, 2022(792):61–91, 2022.
- [DZ19] Alexander Dunn and Alexandru Zaharescu. Sums of Kloosterman sums over primes in an arithmetic progression. *The Quarterly Journal of Mathematics*, 70(1):319–342, 2019.
- [Eke07] Torsten Ekedahl. *On The Adic Formalism*, pages 197–218. Birkhäuser Boston, Boston, MA, 2007.
- [ELS20] Jordan S. Ellenberg, Wanlin Li, and Mark Shusterman. Nonvanishing of hyperelliptic zeta functions over finite fields. *Algebra & Number Theory*, 14(7):1895–1909, 8 2020.
- [EVW16] Jordan S. Ellenberg, Akshay Venkatesh, and Craig Westerland. Homology stability for Hurwitz spaces and the Cohen-Lenstra conjecture over function fields. *Annals of Mathematics*, 183:729–786, 2016.
- [FFK25] Arthur Forey, Javier Fresán, and Emmanuel Kowalski. Arithmetic fourier transforms over finite fields: generic vanishing, convolution, and equidistribution, 2025.
- [FK12] Sergey Finashin and Viatcheslav Kharlamov. Abundance of Real Lines on Real Projective Hypersurfaces. *International Mathematics Research Notices*, 2013(16):3639–3646, 06 2012.

- [FLR23] Tony Feng, Aaron Landesman, and Eric M. Rains. The geometric distribution of Selmer groups of elliptic curves over function fields. *Mathematische Annalen*, 387:615–687, 2023.
- [Fu15] Lei Fu. *Etale Cohomology Theory*, volume 14 of *Nankai Tracts in Mathematics*. World Scientific, 2015.
- [FvdG04] Carel Faber and Gerard van der Geer. Complete subvarieties of moduli spaces and the Prym map. *Journal für die reine und angewandte Mathematik*, 2004(573):117–137, 2004.
- [GL96] Ofer Gabber and François Loeser. Faisceaux pervers ℓ -adiques sur un tore. *Duke Math J.*, 83(3):501–606, 1996.
- [Gol06] Dorian Goldfeld. Conjectures on elliptic curves over quadratic fields. In *Number Theory Carbon-dale 1979: Proceedings of the Southern Illinois Number Theory Conference Carbondale, March 30 and 31, 1979*, pages 108–118. Springer, 2006.
- [GR04] Alexander Grothendieck and Michèle Raynaud. Revêtements étales et groupe fondamental (SGA 1). eprint arXiv:matyh/0206203, 2004. Updated edition of the book of the same title published by Springer-Verlag in 1971 as volume 224 of the series Lecture Notes in Mathematics.
- [Gro77] Alexander Grothendieck. *Cohomologie l -adique et fonctions L* Séminaire de Géométrie Algébrique due Bois-Marie 1965-1966 (SGA 5), volume 589 of *Springer Lecture Notes*. Springer-Verlag, 1977. Avec la collaboration de I. Bucur, C. Houzel, L. Illusie, J.-P. Jouanolou, et J.-P. Serre.
- [GSDV72] Alexander Grothendieck, Bernad Saint-Donat, and Jean-Louis Verdier. *Theorie des Topos et Cohomologie Etale des Schemas. Seminaire de Geometrie Algebrique du Bois-Marie 1963-1964 (SGA 4)*. Lecture Notes in mathematics. Springer-Verlag Berlin Heidelberg, 1 edition, 1972.
- [GV72] Alexander Grothendieck and Jean-Louis Verdier. *Theorie des Topos et Cohomologie Etale des Schemas. Seminaire de Geometrie Algebrique du Bois-Marie 1963-1964 (SGA 4)*. Lecture Notes in mathematics. Springer-Verlag Berlin Heidelberg, 1 edition, 1972.
- [HB04] David Heath-Brown. The average analytic rank of elliptic curves. *Duke Math. J.*, 122:591–623, 2004.
- [HK25] Chris Hall and Hyun Jong Kim. Independence of ℓ (title to be determined). In progress, 2025.
- [HM73] Dale Husemoller and John Milnor. *Symmetric Bilinear Forms*, volume 73 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 2. Folge*. Springer Berlin Heidelberg, 1973.
- [HSBT05] Michael Harris, Nick Shepherd-Barron, and Richard Taylor. Ihara’s lemma and potential automorphy, 2005.
- [Hub97] Annette Huber. Mixed perverse sheaves for schemes over number fields. *Compositio Mathematica*, 108:107–121, 1997.
- [Ibu22] Tomoyoshi Ibukiyama. Supersingular loci of low dimensions and parahoric subgroups. *Osaka Journal of Mathematics*, 59:703–726, 2022.
- [Jor05] Andrei Jorza. The birch and swinnerton-dyer conjecture for abelian varieties over number fields, 2005.
- [Joy02] André Joyal. Quasi-categories and kan complexes. *Journal of Pure and Applied Algebra*, 175(1-3):207–222, 2002.
- [Kat90] Nicholas M. Katz. *Exponential sums and differential equations*, volume 124 of *Annals of Mathematics Studies*. Princeton University Press, 1990.
- [Kat96] Nicholas M. Katz. *Rigid Local Systems*, volume 139 of *annals of Mathematics Studies*. Princeton University Press, 1996.
- [Kat98] Nicholas M. Katz. *Gauss Sums, Kloosterman Sums, and Monodromy Groups*, volume 116 of *Annals of Mathematics Studies*. Princeton University Press, 1998.
- [Kat12] Nicholas M. Katz. *Convolution and Equidistribution Sato-Tate Theorems for Finite-Field Mellin Transforms*, volume 180 of *Annals of Mathematics Studies*. Princeton University Press, 2012.
- [Kim23] Hyun Jong Kim. `trouver`, 2023. GitHub repository: <https://github.com/hyunjongkimmath/trouver>.
- [Kim24] Hyun Jong Kim. *Cohen-Lenstra heuristics and vanishing of zeta functions for superelliptic curves over finite fields*. PhD thesis, University of Wisconsin-Madison, 2024.
- [KL85] Nicholas M. Katz and Gérard Laumon. Transformation de fourier et majoration de sommes exponentielles. *Publications Mathématiques de l’IHÉS*, 62:145–202, 1985.

- [KLSW23] Jesse Leo Kass, Marc Levine, Jake P. Solomon, and Kirsten Wickelgren. A quadratically enriched count of rational curves. arXiv 2307.01936, 2023.
- [KM23] Seoyoung Kim and M. Ram Murty. From the Birch and Swinnerton-Dyer conjecture to Nagao’s conjecture. *Mathematics of Computation*, 92(339):385–408, 2023.
- [KMS17] Emmanuel Kowalski, Philippe Michel, and Will Sawin. Bilinear forms with Kloosterman sums and applications. *Annals of Mathematics*, 186:413–500, 2017.
- [Knu91] Max-Albert Knus. *Quadratic and Hermitian Forms over Rings*, volume 294 of *Grundlehren der mathematischen Wissenschaften*. Springer Berlin, 1991.
- [KP24] Hyun Jong Kim and Sun Woo Park. Global A^1 degrees of covering maps between modular curves, 2024.
- [Krä14] Thomas Krämer. Perverse sheaves on semiabelian varieties. *Rendiconti del Seminario Matematico della Università di Padova*, 132:83–102, 2014.
- [KS99] Nicholas M. Katz and Peter Sarnak. *Random matrices, Frobenius eigenvalues, and monodromy*, volume 45. American Mathematical Society, 1999.
- [KS06] Masaki Kashiwara and Pierre Schapira. *Categories and Sheaves*, volume 332 of *Grundlehren der mathematischen Wissenschaften*. Springer Berlin, Heidelberg, 1 edition, 2006.
- [KS22] Timo Keller and Michael Stoll. Exact verification of the strong bsd conjecture for some absolutely simple abelian surfaces. *Comptes Rendus Mathématique*, 360:483–489, 2022.
- [KW13] Reinhardt Kiehl and Rainer Weissauer. *Weil Conjectures, Perverse Sheaves and ℓ -adic Fourier Transform*, volume 42 of *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge / A Series of Modern Surveys in Mathematics*. Springer Berlin, Heidelberg, 2013.
- [KW15] Thomas Krämer and Rainer Weissauer. Vanishing theorems for constructible sheaves on abelian varieties. *J. Algebraic Geometry*, 24:531–568, 2015.
- [KW19] Jesse Leo Kass and Kirsten Wickelgren. The class of Eisenbud-Khimshiashvili-Levine is the local A^1 – Brouwer degree. *Duke Mathematical Journal*, 168(3):429–469, 2019.
- [KW21] Jesse Leo Kass and Kirsten Wickelgren. An arithmetic count of the lines on a smooth cubic surface. *Compositio Mathematica*, 157(4):677–709, 2021.
- [Laf02] Laurent Lafforgue. Chtoucas de Drinfeld et correspondance de Langlands. *Inventiones mathematicae*, 147:1–241, 2002.
- [Lom17] Davide Lombardo. Galois representations attached to abelian varieties of cm type. *Bulletin de la Société mathématique de France*, 145(3):469–501, 2017.
- [Lur09] Jacob Lurie. *Higher topos theory*. Princeton University Press, 2009.
- [May99] Jon Peter May. *A Concise Course in Algebraic Topology*. Chicago Lectures in Mathematics. University of Chicago Press, 1999.
- [Mil80] James S. Milne. *Etale cohomology*. Number 33 in Princeton Mathematical Series. Princeton university press, 1980.
- [Mil07] James S. Milne. Quotients of Tannakian categories. *Theory and Applications of Categories*, 18(21):654–664, 2007.
- [Mil13] James S. Milne. Lie algebras, algebraic groups, and lie groups, 2013. Available at www.jmilne.org/math/.
- [Mil17] James S. Milne. *Algebraic Groups*, volume 170 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2017.
- [Mit14] Howard H. Mitchell. The subgroups of the quaternary abelian linear group. *Trans. Amer. Math. Soc.*, 15(4):379–396, 1914.
- [Mor06] Fabien Morel. A^1 -algebraic topology. In *International Congress of Mathematicians*, volume 2, pages 1035–1059, 2006.
- [Mor12] Fabien Morel. *A^1 -Algebraic Topology over a field*. Lecture Notes in Mathematics. Springer Berlin, Heidelberg, 2012.
- [MV99] Fabien Morel and Vladimir Voevodsky. A^1 -homotopy theory of schemes. *Publications Mathématiques de l’IHÉS*, 90:45–143, 1999.
- [Nag97] Koh-ichi Nagao. $Q(t)$ -rank of elliptic curves and certain limit coming from the local points. *Manuscripta mathematica*, 92(1):13–32, 1997.

- [Nee02] Amnon Neeman. A counterexample to a 1961 "theorem" in homological algebra. *Inventiones Mathematicae*, 148(2):397–420, 2002. With an appendix by P. Deligne. Provides the counterexample to Roos’s 1961 result.
- [nLa25a] nLab authors. geometric morphism. <https://ncatlab.org/nlab/show/geometric+morphism>, July 2025. Revision 61.
- [nLa25b] nLab authors. homotopy group of a spectrum. <https://ncatlab.org/nlab/show/homotopy+group+of+a+spectrum>, June 2025. Revision 7.
- [nLa25c] nLab authors. Introduction to Stable homotopy theory – 1-1. <https://ncatlab.org/nlab/show/Introduction+to+Stable+homotopy+theory+--+1-1>, June 2025. Revision 43.
- [nLa25d] nLab authors. model structure on topological sequential spectra. <https://ncatlab.org/nlab/show/model+structure+on+topological+sequential+spectra>, June 2025. Revision 61.
- [nLa25e] nLab authors. point of a topos. <https://ncatlab.org/nlab/show/point+of+a+topos>, July 2025. Revision 53.
- [nLa25f] nLab authors. sheafification. <https://ncatlab.org/nlab/show/sheafification>, September 2025. Revision 40.
- [nLa25g] nLab authors. stable homotopy category. <https://ncatlab.org/nlab/show/stable+homotopy+category>, June 2025. Revision 31.
- [OT14] Christian Okonek and Andrei Teleman. Intrinsic signs and lower bounds in real algebraic geometry. *Journal für die reine und angewandte Mathematik (Crelles Journal)*, 2014(688):219–241, 2014.
- [Poo18] Bjorn Poonen. Heuristics for the arithmetic of elliptic curves. In *Proceedings of the International Congress of Mathematicians: Rio de Janeiro 2018*, pages 399–414. World Scientific, 2018.
- [PR12] BJORN POONEN and ERIC RAINS. Random maximal isotropic subspaces and selmer groups. *Journal of the American Mathematical Society*, 25(1):245–269, 2012.
- [Pri24] Rachel Pries. The torelli locus and newton polygons, 2 2024. Lecture Notes for the 2024 Arizona Winter School.
- [Pri25] Rachel Pries. Some cases of oort’s conjecture about newton polygons of curves. *Nagoya Mathematical Journal*, 257:93–103, 2025.
- [PW21] Sabrina Pauli and Kirsten Wickelgren. Applications to $\mathbb{A}^1$ -enumerative geometry of the $\mathbb{A}^1$ -degree. *Research in the Mathematical Sciences*, 8(24):24–29, 2021.
- [Roo61] Jan-Erik Roos. Sur les foncteurs dérivés de $\varprojlim$. applications. *Comptes Rendus de l’Académie des Sciences, Paris*, 252:3702–3704, 1961. The original claim regarding (AB4*) categories.
- [Roo06] Jan-Erik Roos. Derived functors of inverse limits revisited. *Journal of the London Mathematical Society*, 73(1):65–83, 2006. The formal correction requiring the category to be Grothendieck.
- [Ros02] Michael Rosen. *Number Theory in Function Fields*, volume 210 of *Graduate Texts in Mathematics*. Springer-Verlag New York, 2002.
- [RS98] Michael Rosen and Joseph H. Silverman. On the rank of an elliptic surface. *Inventiones mathematicae*, 133:43–67, 1998.
- [Rud87] Walter Rudin. *Real and Complex Analysis*. Mathematics Series. McGraw-Hill Book Company, 3 edition, 1987.
- [Saw24] Will Sawin. General multiple dirichlet series from perverse sheaves. *Journal of Number Theory*, 262:408–453, 2024.
- [Ser72] Jean-Pierre Serre. Propriétés galoisiennes des points d’ordre fini des courbes elliptiques. *Invent. math.*, 15:259–331, 1972.
- [Ser00] Jean-Pierre Serre. Lettre à marie-france vignéras du 10/2/1986. *Oeuvres–Collected Papers*, 4:38–55, 2000.
- [Ser03] Christian Serpé. Resolution of unbounded complexes in grothendieck categories. *Journal of Pure and Applied Algebra*, 177(1):103–112, 2003.
- [SFFK23] Will Sawin, Arthur Forey, Javier Fresán, and Emmanuel Kowalski. Quantitative sheaf theory. *Journal of the American Mathematical Society*, 36(3):653–726, 2023.
- [Sil89] Joseph H. Silverman. Elliptic curves of bounded degree and height. *Proceedings of the American Mathematical Society*, 105(3):540–545, 1989.

- [Sil09] Joseph H. Silverman. *The arithmetic of elliptic curves*, volume 106 of *Graduate Texts in Mathematics*. Springer-Verlag, 2 edition, 2009.
- [Spa88] Nicolas Spaltenstein. Resolutions of unbounded complexes. *Compositio Mathematica*, 65(2):121–154, 1988.
- [ST21] Will Sawin and Jacob Tsimmerman. Bounds for the stalks of perverse sheaves in characteristic p and a conjecture of shende and tsimmerman. *Inventiones mathematicae*, 224(1):1–32, 2021.
- [Sta25] The Stacks project authors. The stacks project. <https://stacks.math.columbia.edu>, 2025.
- [Tat65] John T. Tate. Algebraic cycles and poles of zeta functions. In *Arithmetical Algebraic Geometry (Proc. Conf. Purdue Univ., 1963)*, pages 93–110. Harper & Row, 1965. Also in *Collected works of John Tate* (2 vols.), Amer. Math. Soc. (2016), vol. 2.
- [Tat66] John Tate. On the conjectures of Birch and Swinnerton-Dyer and a geometric analog. In *Séminaire Bourbaki : années 1964/65 1965/66, exposés 277-312*, number 9 in *Astérisque*, pages 415–440. Société mathématique de France, 1966. talk:306.
- [Tay08] Richard Taylor. Automorphy for some l -adic lifts of automorphic mod l galois representations. ii. *Publications mathématiques*, 108:183–239, 2008.
- [Voe98] Vladimir Voevodsky. A_1 -homotopy theory. In *Proceedings of the international congress of mathematicians*, volume 1, pages 579–604. Berlin, 1998.
- [Wei94] Charles A. Weibel. *An Introduction to Homological Algebra*, volume 38 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 1994. First paperback edition 1995 Reprinted 1997.
- [Wik25] Wikipedia contributors. Frobenius endomorphisms#frobenius for schemes — Wikipedia, the free encyclopedia, 2025. [Online; accessed 08-July-2025].
- [You06] Matthew Young. Low-lying zeros of families of elliptic curves. *Journal of the American Mathematical Society*, 19(1):205–250, 2006.
- [Yu97] Jiu-Kang Yu. Toward a proof of the Cohen-Lenstra conjecture in the function field case. preprint, 1997.